ECE: Advanced Information and Network Security
Quiz 1
Spring 2005

1. Suppose you encrypt using an affine cipher, then encrypt the encryption using another affine cipher (both are working mod 26). Is there any advantage to doing this, rather than using a single affine cipher? Why or why not?

2. Suppose we modify the Feistel setup as follows. Divide the plaintext into three equal blocks: L_0, M_0, R_0 (left, middle, right). Let the key for the ith round be K_i and let f be some function that produces an appropriate size output. The ith round of encryption is given by $L_i = R_{i-1}$, $M_i = L_{i-1}$, $R_i = f(K_i, R_{i-1}) \oplus M_{i-1}$. This continues for n rounds. Consider the decryption algorithm that starts with the ciphertext A_n, B_n, C_n and uses the algorithm $A_{i-1} = B_i$, $B_{i-1} = f(K_i, A_i) \oplus C_i$, $C_{i-1} = A_i$. This continues for n rounds, down to A_0, B_0, C_0. Show that $A_i = L_i$, $B_i = M_i$, $C_i = R_i$ for all i, so that the decryption algorithm returns the ciphertext.